Labyrinthos

Rescue Simulation

from Taiwan

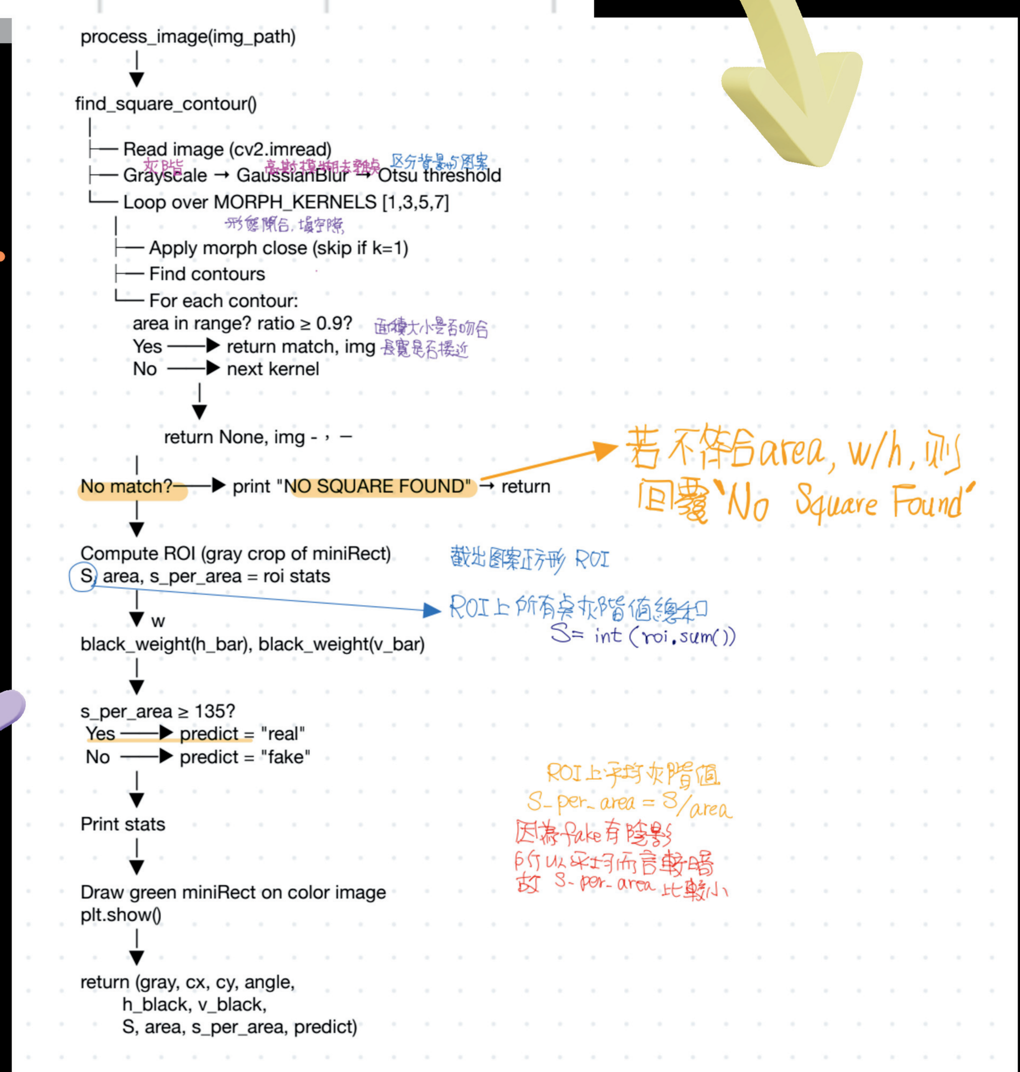
- Yin Shao Chi (17y)
- Tester，coding
- Experienced in competitive robotics, specifically with
- Arduino and VEX V5 platforms

- Kao Chen wei (15y)
- Engineering Journal，coding
- Early exposure to programming，arduino , IQ, Vex v5
- Domestic and international competitions

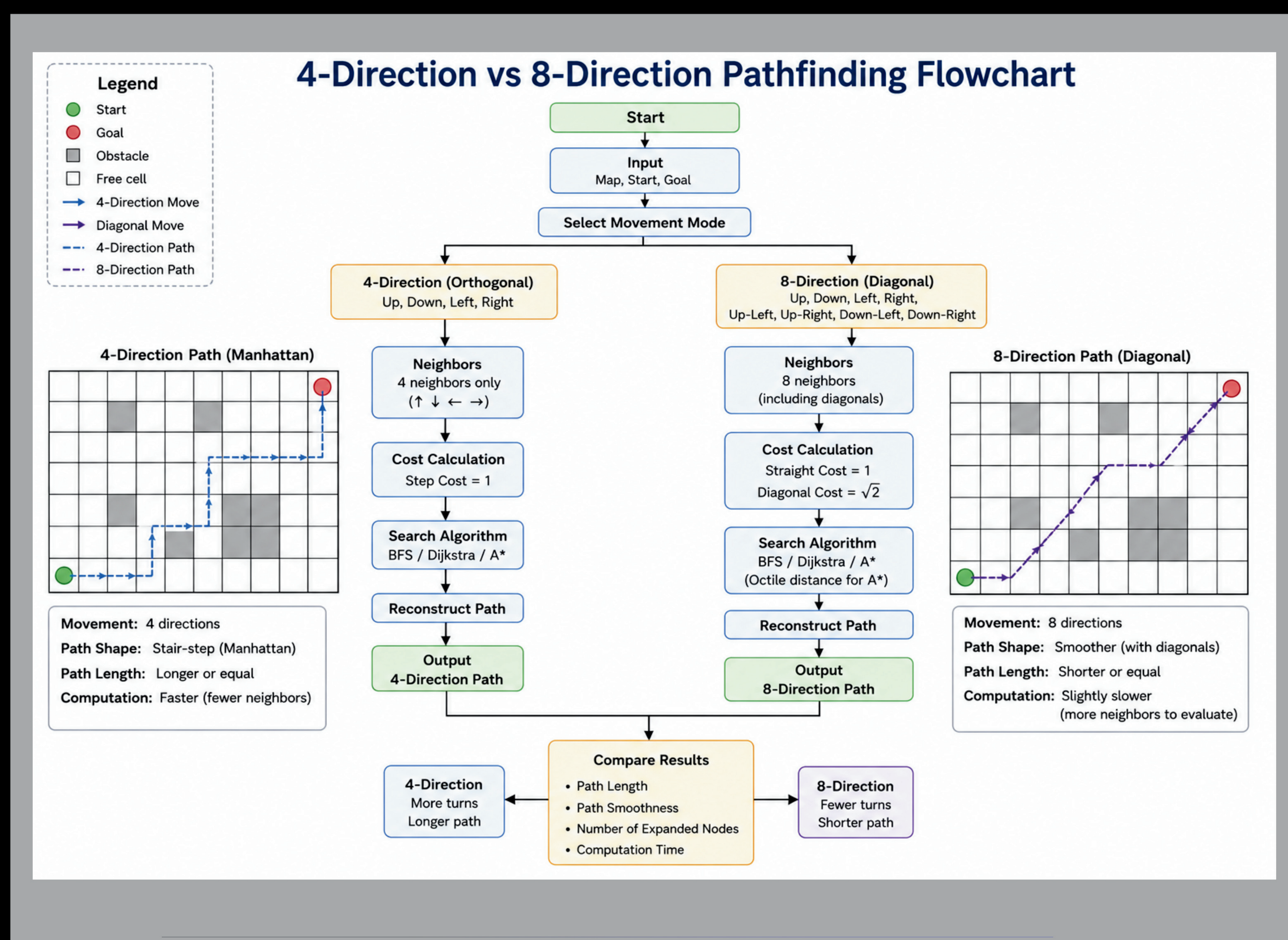
- Chen Yu Wei (16y)
- Engineering Journal，coding
- Early exposure to programming
- arduino, Ev3,Vex v5 Domestic and international competitions

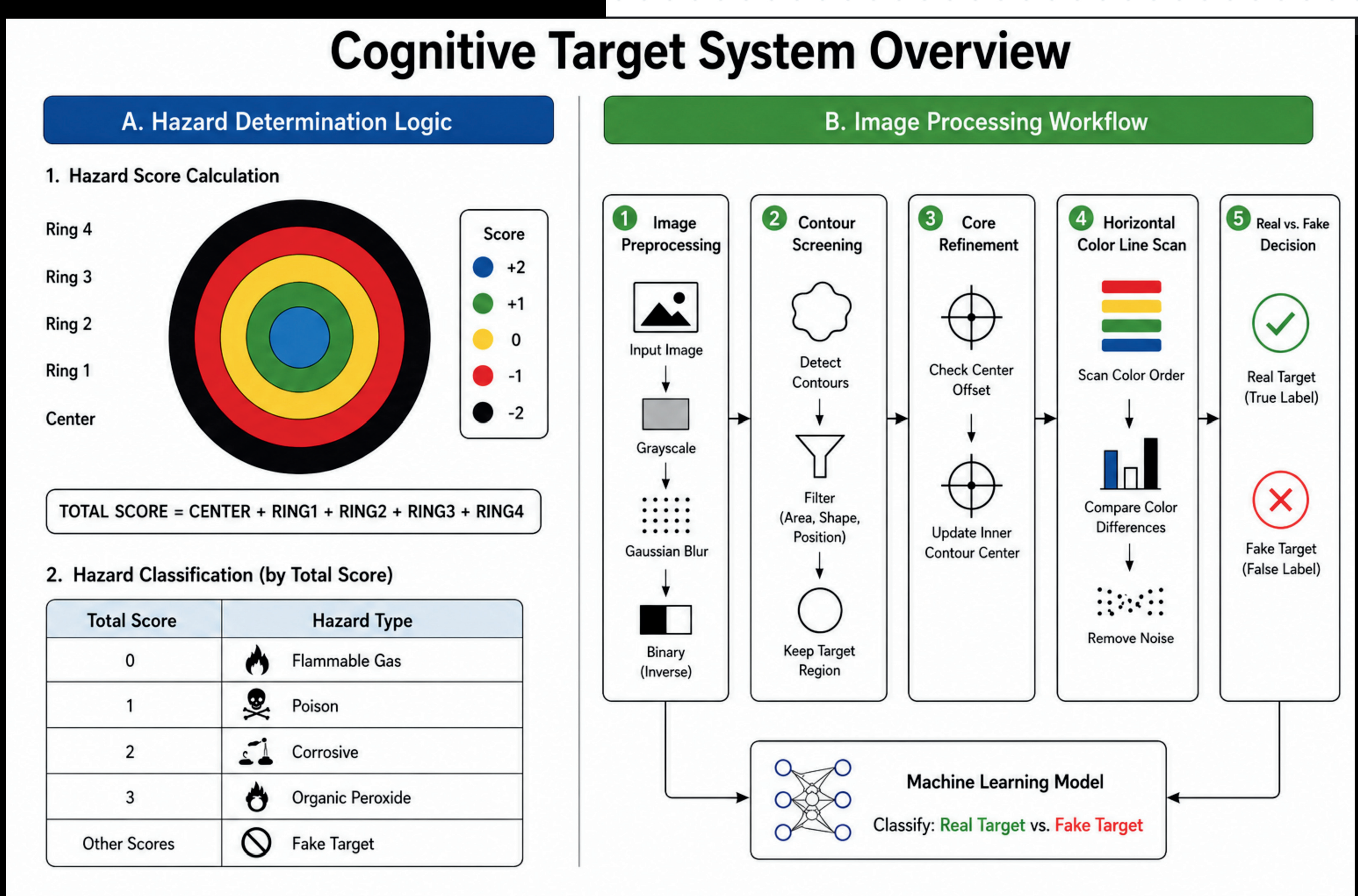
	A	B	C	D	E	F
1	filename	area	aspect_ratio	morph_k	angle	cx
2	omegaF1.jpg	372	0.917	3	-90	30
3	omegaF2.jpg	576	1	1	-90	29
4	omegaF3.jpg	263.5	1	5	-90	42.5
5	omegaF4.jpg	332.5	1	5	-90	36
6	omegaF5.jpg	288	0.95	7	-90	18.5
7	phiF1.jpg	326	0.95	7	-90	53
8	phiF2.jpg	377	1	5	-90	34
9	phiF3.jpg	313	1	3	-90	15
10	phiF4.jpg	321	0.96	3	-90	41.5
11	psiF1.jpg	291	0.958	3	-90	13.5
12	psiF2.jpg	312	0.958	1	-90	26.5
13	psiF3.jpg	281	1	5	-90	49

We found that grayscale values can be used to highlight colors.



4 Distances(F/R/FR/FL) :Distance Detection for Obstacle Avoidance
1 Color sensor : Floor color marker detection
2 Camara(R/F) : Object detection for scoring
1 Lidar :Enhancing positional accuracy by matching LiDAR scans against the known field layout
1 Inertial_unit : Maintaining a constant heading during autonomous navigation to prevent drifting
1 GPS:Geofencing & Area Marking





Test scores

Image recognition accuracy

95% up

Map Coverage Percentage

85% up

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